

India's #1 Website For Electronics Engineers Working Across The Globe

Established: **1998**
Registered Users: **300,000+**



Monthly Statistics

Page Impressions:
750,000+

Unique Techies:
150,000



electronicsforu.com
If it's electronics, it's here

For any queries, please contact
our team at efyenc@efyindia.com
OR +91-11-26810601

an
EFYGROUP
PUBLICATION

nate path within MLLN.

With these network elements, the MLLN is able to provide functions such as fault, configuration, accounting, performance and security management.

Network management system (NMS). It is built on open architecture and relational database system and manages all the network elements and their functions centrally. It is able to configure, provision, manage and monitor all aspects and parameters of the remote elements of the MLLN network centrally without the need of any local intervention.

On-demand bandwidth configuration is performed at NMS and priority can also be set for a particular leased line. This enables the high-priority customer lines to be routed first to the standby route in case of failure of the main route.

Any change of configuration of any network element is auto-recognised by the NMS. NMS is also capable of re-initialisation of the network element in the event of software/hardware failure. A predefined routing schedule is used by NMS to perform fast re-establishment of circuits within the network across alternative paths totally automatically, in the event of failure.

A detailed fault report is generated in order to identify the exact problem so that immediate corrective measures can be taken in order to restore the services.

The fault information provided by NMS contains type of faulty network element, the time at which fault occurred, time when it corrected, etc. The NMS keeps polling all the network elements after some predetermined interval and generates alarms so that proactive measures can be taken.

Digital cross connect (DXC). It is a large-capacity cross-connect device that separates channels coming from other devices and rearranges them into new channels for output. A digital cross-connection allows lower-level time division multiplexing (TDM) bit streams to be rearranged and inter-connected amongst higher-level TDM sig-

nals. The signal is first de-multiplexed into a lower level after which it is cross-connected and then multiplexed again.

Versatile multiplexer (VMUX). It is a small capacity cross connect device with several channel interfaces which is installed at different sites for providing user connectivity. VMUX multiplexes all tributaries coming from other devices and forms a higher hierarchy level output at the specified port. The VMUX is provided with different types of interfaces to connect SDSL and HDSL product family modems. The number of interfaces (such as 64/128kbps, n x 64kbps, E1 or hotline) depends on the type of VMUX configuration (VMUX type I, VMUX type II, VMUX type III/DC operation and VMUX type III AC operation).

Network terminating unit (NTU). It is simply a base band modem and which is located at the customer's premises. The NTU works on normal AC supply. NTUs of various capacities (64/128kbps and n x 64kbps) are available with several interfaces (V.35, G.703, Ethernet). NTU also allows for the use of existing telecom copper cables (twisted pair) for digital traffic with medium distances (~5 km) and high speeds.

Conclusion

In the changing economic environment, dependence of organisational and industrial activities on leased circuits is increasing. In such a scenario, high QoS, high efficiency, highly secured network, customer-oriented tariffs along with desired bandwidth, time-dependent bandwidth provisioning, no congestion, centralised control and monitoring, lower lead time for new installations and proactive fault maintenance prove MLLN to be a commercial boon for corporate and individual customers. ●

The author is working with Bharat Sanchar Nigam Limited. He holds a Ph.D. degree in electronics engineering from Indian Institute of Technology (BHU), Varanasi. His current research interests include wired and wireless technologies for high-speed Internet access